

PATIENT COMPLEXITY AND PATIENT NEEDS

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The concept of *patient complexity* is intuitive to most primary care clinicians, but difficult to define in an agreed precise manner [1]. Often complexity is linked to an arbitrary - if significant - outcome such as hospital admission risk, based predominantly on medical risk factors [2, 3]. However, clinicians typically experience complexity as consultations that are multi-faceted, time-consuming, shaped by psychological, socio-economic and systemic factors and with multiple - sometimes problematic - interactions between patient, multiple health care providers and community [1, 4, 5]. Some define complexity as the *gap between an individual's needs and the capacity of the healthcare system to answer those needs* [6]. Greater complexity often corresponds with higher overall healthcare use and social support needs [1].

Several reporting tools for general practice electronic medical records (EMRs) measure patient complexity for reporting to Primary Health Networks (PHNs). The PHN Performance and Quality Framework includes indicator 'P12', which tracks potentially preventable hospitalisation [7].

- **PenCS** incorporates a CSIRO developed tool using logistic regression to predict hospitalisation using Australian general practice EMR data. Tool workings and development is well-documented and transparent [8]. The authors are happy to share the methodology *exactly* for non-profit purposes. It may be possible to adapt the tool to other outcomes.
- **POLAR** uses the *Diversion 2* tool, developed using machine learning techniques with demographic, diagnoses, medication, pathology and measurement data extracted from general practices with GRHANITE and linked with hospital admission data [9, 10].
- **Primary Sense** integrates the extensively developed, international and proprietary **John Hopkins ACG System** [11, 12]. The John Hopkins ACG system has been used to estimate current and likely future care needs in primary care, including more equitable funding of general practices in the NHS [13, 14]. However, the use of John Hopkins ACG in Primary Sense is limited to the risk of hospital admission. John Hopkins ACG uses *patient segmentation* to group patients into relatively homogenous groups with similar care needs based on medical and socio-economic characteristics [15–17] (Figure 1).

Opportunities

- Patient segmentation - with social need and other modifiers for special care needs - has potential value in primary care settings to predict and provide services tailored towards behavioural, clinical or demographic drivers of health over and above predictors based on a single score [15, 17–19] (Figure 2). The patient segmentation approach has been adapted for diverse populations, including in Southeast Asia [20] (Figures 3, 4).
- Predictive models for specific outcomes within patient segments might improve predictive power, as both patient needs and importance of specific risk-factors varies between patient segments [21].
- Alternative intermediate outcomes, e.g. primary care visit frequency and duration or correspondence volume, could be explored [5].

Figures

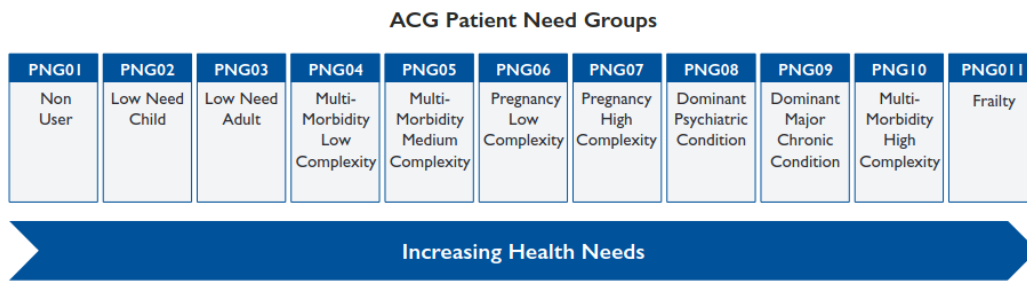


Figure 1: John Hopkins ACG Patient Needs Groups

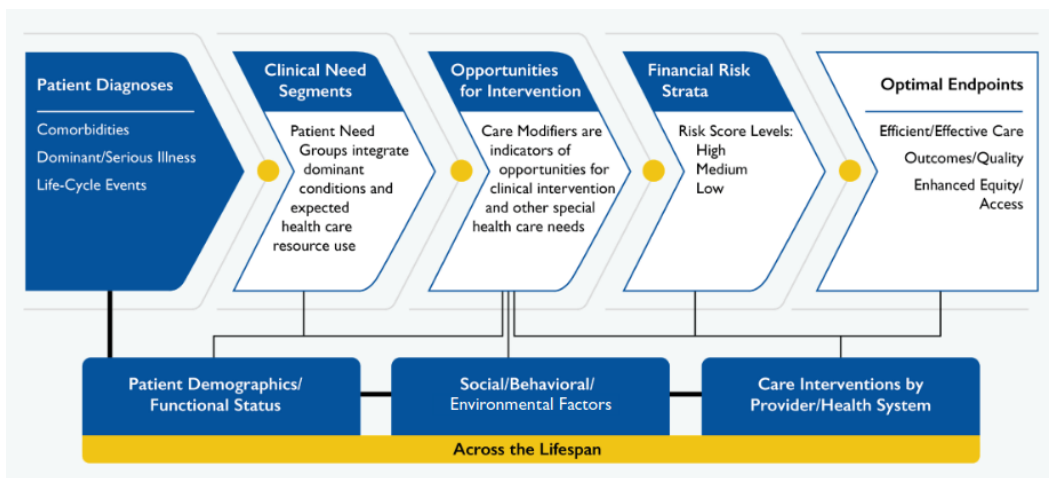


Figure 2: John Hopkins ACG care modifiers

Broad Groups	Children & Adolescence 18 years and below	Adults 19 - 64 years	Seniors 65 years and above
Staying Healthy No Major Health Needs (Preventive Care)	[A] Mostly Healthy Children, Adults and Seniors With no known chronic condition and no known clinical and social risk factor. Maintenance of healthy lifestyle, age appropriate screening, age appropriate preventive health services (e.g. vaccination).		
	[B] At Risk Children and Adults With no known chronic condition and with one clinical or social risk factor such as low socio-economic status and obesity) Targeted screening of at risk population (e.g. genetic/familial diseases), targeted interventions for at risk population.	[C] At Risk Seniors With no chronic condition and one clinical or social risk factor. Interventions to address risk factors to prevent disease.	
Living Well with Illness With Chronic Disease (Chronic Disease Management)	[D] Children with Chronic Condition Individuals who have chronic condition and require minimal follow-up at acute care.	[E] Adults and Seniors with Simple Chronic Condition With condition(s) and case-mix which can be managed primarily in primary care with minimal support from specialist care.	
		[F] Adults and Seniors with Complex Chronic Condition With condition(s) and case-mix which require appropriate co-management between primary and specialist care for chronic disease management.	
		[G] Adults and Seniors with Mental Health Condition With mental health condition. Require primary and specialist care for mental health, with support in the community.	
		[H] Adults and Seniors with Cancer With cancer. Require specialist care with support in the community.	
Getting Well Time-Limited Health Needs (Transitional Care)	[I] Children with Transitional & Long-Term Needs Require additional specialist care to return to baseline function and/or delay progression of disease.	[J] Adults and Seniors with Transitional Needs Had major acute episodic health needs and subsequently require additional support post-discharge and integrate back to the community and return of baseline function.	
Maximising Quality Life Years Frail and End-of-Life (Disease Management and Palliative Care)		[K] Frail Adults and Seniors With diminished function (physical and/or cognitive) or are institutionalised in nursing homes.	
		[L] End-of-Life Children, Adults and Seniors Advanced care planning and accessible end-of-life services to support end-of-life preferences.	

Figure 3: Health Segment Classification of Population Encompassed within Singapore (HealthSCOPES)

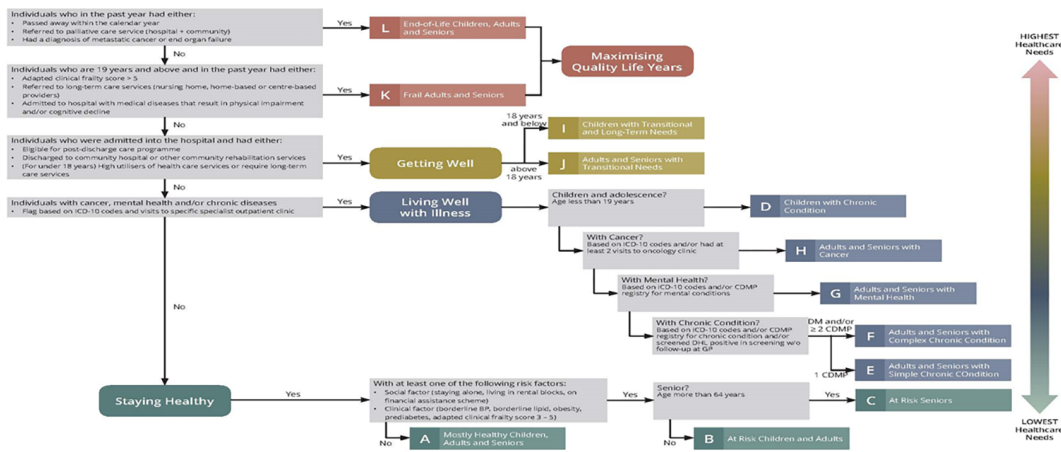


Figure 4: Classification hierarchy of HealthSCOPES, outlining the criteria to categorize an individual into one of the 12 segments

References

- Nicolaus S, Crelier B, Donzé JD, Aubert CE (2022) Definition of patient complexity in adults: A narrative review. *Journal of Multimorbidity and Comorbidity* 12:26335565221081288. <https://doi.org/10.1177/26335565221081288>
- Hudson SM, Mueller M, Hester WH, Magwood GS, Newman SD, Laken MA (2014) At-risk characteristics for hospital admissions and ED visits. *Journal for specialists in pediatric nursing : JSPN* 19(2):183–193. <https://doi.org/10.1111/jspn.12068>
- Islam N, Shabnam S, Khan N, et al (2024) Combinations of multiple long term conditions and risk of hospital admission or death during winter 2021-22 in England: population based cohort study. *BMJ Medicine* 3(1). <https://doi.org/10.1136/bmjmed-2024-001016>
- Salisbury C, Lay-Flurrie S, Bankhead CR, et al (2021) Measuring the complexity of general practice consultations: A delphi and cross-sectional study in english primary care. *The British Journal of General Practice* 71(707):e423–e431. <https://doi.org/10.3399/BJGP.2020.0486>
- Oliveira CA de, Weber B, Santos JLF dos, et al (2022) Health complexity assessment in primary care: A validity and feasibility study of the INTERMED tool. *PLoS ONE* 17(2):e0263702. <https://doi.org/10.1371/journal.pone.0263702>
- Grembowski D, Schaefer J, Johnson KE, et al (2014) A conceptual model of the role of complexity in the care of patients with multiple chronic conditions. *Medical Care* 52:S7. <https://doi.org/10.1097/MLR.000000000000045>
- Department of Health, Commonwealth of Australia (2018) **Primary health networks (PHN) performance and quality framework**
- Khanna S, Rolls DA, Boyle J, et al (2019) A risk stratification tool for hospitalisation in Australia using primary care data. *Scientific Reports* 9(1):5011. <https://doi.org/10.1038/s41598-019-41383-y>

9. South Western Sydney PHN (2023) **POLAR update: New report to calculate patient risk of hospitalisation**
10. Pearce C (2017) **POLAR Diversion: Using GP data to calculate risk of ED presentation**
11. Gold Coast Primary Health Network **Identify patients with high complexity who are missing cycles of care using primary sense**
12. Primary Sense **Primary sense data extraction guide**
13. Shepherd D (2023) **Use the ACG system to develop a new funding model for primary care in england**
14. Shepherd D, Farrar C (2025) **A fairer way of funding general practice - learning from LLR and frimley ICSs**
15. Johns Hopkins ACG System (2022) **Patient need groups**
16. Johns Hopkins ACG System (2022) **Understanding patient segmentation**
17. Johns Hopkins ACG System (2022) **Social need markers - what are they and why are they important?**
18. National Academy of Medicine; The Learning Health System Series (2017) **Effective Care for High-Need Patients: Opportunities for Improving Outcomes, Value, and Health**. National Academies Press (US), Washington (DC)
19. Lynn J, Straube BM, Bell KM, Jencks SF, Kambic RT (2007) Using population segmentation to provide better health care for all: The “bridges to health” model. The Milbank Quarterly 85(2):185–208. <https://doi.org/10.1111/j.1468-0009.2007.00483.x>
20. Ang IYH, Rahman N, Wong SH, et al (2025) Development and validation of the health segment classification of population encompassed within singapore (Health-SCOPES) framework. PLOS ONE 20(1):e0317016. <https://doi.org/10.1371/journal.pone.0317016>
21. Langton JM, Wong ST, Burge F, et al (2020) Population segments as a tool for health care performance reporting: An exploratory study in the canadian province of british columbia. BMC Family Practice 21(1):98. <https://doi.org/10.1186/s12875-020-01141-w>